BOARD QUESTION PAPER: MARCH 2026

Mathematics Part - II

Time: 2 Hours **Max. Marks: 40**

Note:

i. *All* questions are compulsory.
ii. Use of a calculator is not allowed.
iii. The numbers to the right of the questions indicate full marks.
iv. In case of MCQs [Q. No. **1(A)**] only the first attempt will be evaluated and will be given credit.
v. Draw proper figures wherever necessary.
vi. The marks of construction should be clear. Do not erase them.
vii. Diagram is essential for writing the proof of the theorem.

Q.1. (A) For each of the following sub-question four alternative answers are given. Choose the correct alternative and write its alphabet. **[4]**

i. Out of the following which is the Pythagorean triplet?
(A) (1, 5, 10) (B) (3, 4, 5)
(C) (2, 2, 2) (D) (5, 5, 2)

ii. Two circles having radii 4 cm and 5 cm touch each other internally. Then distance between the centres is .....
(A) 4 cm (B) 5 cm
(C) 1 cm (D) 9 cm

iii. Out of the following points ..... lies to the right of the origin on X-axis.
(A) (–2, 0) (B) (0, 2)
(C) (2, 3) (D) (2, 0)

iv. If the side of the cube is 5 cm, then the volume of that cube is .....
(A) 10 cm^3 (B) 100 cm^3
(C) 125 cm^3 (D) 25 cm^3

(B) Solve the following sub-questions: **[4]**

i. The ratio of corresponding sides of similar triangles is 3 : 5, then find the ratio of their areas.

ii. Find the side of a square if its diagonal is $10\sqrt{2}$ cm.

iii. If the measure of minor arc is 70°, then find the measure of corresponding major arc.

iv. Angle made by the line with the positive direction of X-axis is 45°, find the slope of line.

Q.2. (A) Complete the following activities and rewrite it *(any two)*: **[4]**

i. In the following figure, seg PS is a tangent segment, line P is a secant. If PQ = 3.6, QR = 6.4, find PS.

Activity:

$\therefore$ $PS^2 = \square \times PR$ (Tangent secant segment theorem)

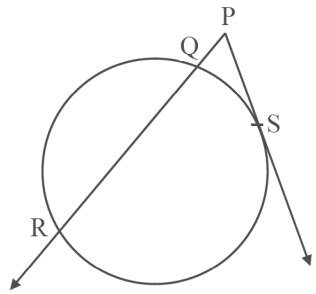
$= PQ \times [PQ + QR]$

$= \square \times [3.6 + 6.4]$

$= 3.6 \times \square$

$\therefore$ $PS^2 = 36$

$PS = \square$



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ii. If $\sec\theta = \frac{25}{7}$, find the value of $\tan\theta$.

Solution:

$1 + \tan^2\theta = \square$

$\therefore \quad 1 + \tan^2 \quad = \square^2$

$\therefore \quad \tan^2\theta = \frac{625}{49} - \square$

$= \frac{625 - 49}{49}$

$\tan^2\theta = \frac{576}{49}$

$\tan\theta = \square$

iii. In the following figure, side of square ABCD is 7 cm. with centre D and radius DA, sector D-AXC is drawn, Fill in the following boxes properly and find out the area of shaded region :

Solution:

Area of square = $(\text{side})^2$

$= (7)^2$

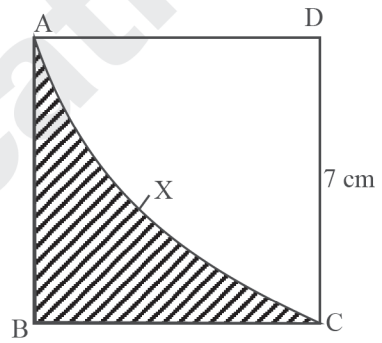
Area of square = $\square$ cm^2

If area of sector (D-AXC) = 38.5 cm^2, then

A (shaded region) = A $\square$ – A $\square$

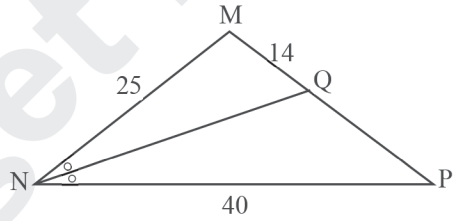
= 49 cm^2 – 38.5 cm^2

= $\square$ cm^2



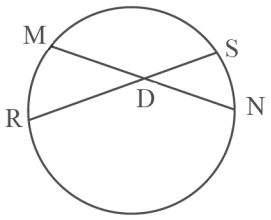
(B) Solve the following sub-questions (any *four*): **[8]**

i. Find QP using given information in the figure :



ii. In ΔRST, $\angle S = 90°$, $\angle T = 30°$, RT = 12 cm, then find RS and ST.

iii. In the following figure, chord MN and chord RS intersect at point D. If RD = 15, DS = 4, MD = 8, find DN.



iv. Find the co-ordinates of the midpoint of the line segment joining points P(0, 6) and Q(12, 20).

v. Prove that: $\sin^2\theta + \cos^2\theta = 1$.

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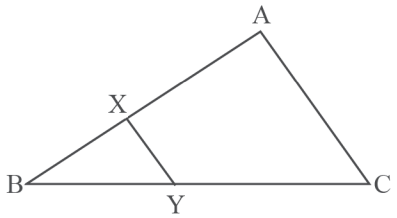
Q.3. (A) Complete the following activities and rewrite it *(any one)*: **[3]**

i. In the following figure, seg XY || side AC. If 2AX = 3BX and XY = 9, complete the activity to find the value of AC.



Activity:

2AX = 3BX

$\therefore \quad \frac{AX}{BX} = \frac{3}{2}$

$\therefore \quad \frac{AX + BX}{BX} = \frac{\square + \square}{2}$ …[By componendo]

$\frac{AB}{BX} = \frac{\square}{2}$

$\Delta BCA \sim \Delta BYX$ …[$\square$ test of similarity]

$\therefore \quad \frac{BA}{BX} = \frac{AC}{XY}$ …[corresponding sides of similar triangles]

$\therefore \quad \frac{5}{\square} = \frac{AC}{9}$

$\therefore \quad AC = \square$

ii. For finding AB and BC with the help of information given in the following figure, complete the following activity:

Activity:

AB = BC …… $\square$

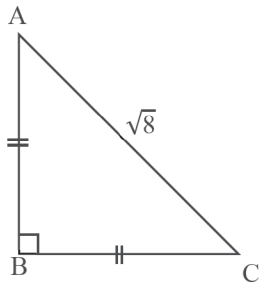
$\therefore \quad \angle BAC = \square$

$\therefore \quad AB = BC = \square \times AC$

$= \frac{1}{\sqrt{2}} \times \square$

$= \frac{1}{\sqrt{2}} \times 2 \times \square$

$AB = BC = \square$



(B) Solve the following sub-questions (any *two*) : **[6]**

i. Prove that : The tangent segment drawn from an external point to the circle are congruent.

ii. Draw a circle with centre O and radius 3.5 cm, take a point P at a distance 5.7 cm from the centre. Draw tangents to the circle from point P.

iii. Find the co-ordinates of point P, if P divides the line segment joining the points A(–1, 7) and B(4, –3) in the ratio 2 : 3.

iv. A bucket is frustum shaped. Its height is 28 cm. Radii of circular faces are 12 cm and 15 cm. Find the capacity of the bucket in litres. $\left(\pi = \frac{22}{7}\right)$

Q.4. Solve the following sub-questions (any *two*) **[8]**

i. $\Delta ABC \sim \Delta ADE$,

In ΔABC, AB = 6.3 cm, $\angle CAB = 50°$, AC = 5.6 cm and $\frac{AB}{AD} = \frac{7}{5}$.

Construct ΔADE.

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ii. Surface area of sphere and total surface area of cube are same. Show that the ratio of the volume of the sphere to that of the cube is $\sqrt{6} : \sqrt{\pi}$.

iii. In Δ ABC, A-X-B and A-Y-C is such that segment XY || side BC.
Segment XY divides Δ ABC in two equal areas, then find $\frac{BX}{AB}$.

Q.5. Solve the following sub-questions (any *two*) : **[3]**

i. Seg PQ is a chord of a circle. Arc PXQ is a major arc and Arc PYQ is a minor arc. $\angle POQ$ is central angle of the circle :

a. Draw the figure from given information.

b. Find the measure of $\angle PXQ$, $\angle PYQ$ and add m $\angle PXQ$, m $\angle PYQ$

c. Find the measure of $\angle POQ$. Write the relation between $\angle PXQ$ and $\angle POQ$.

ii. Two eagles are flying in the sky at height of $10\sqrt{3}$ m above the ground. A boy standing at a point on the ground looks towards the eagles making angles of elevation of 60° and 30° respectively. Find the distance between these two eagles.

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